“ SSH KEY ”

- **What is SSH Key?**

  It means a Secure Shell Key which is simply the “digital fingerprint” of your device. Instead of typing your username and password (or token) every time you upload (push) or download (pull), SSH lets your device and the server to communicate to each other and verify identity automatically and securely.

- **What does SSH Key consist of?**

  1. **Public Key:**

     This is what you publish everywhere, and upload it to your GitHub account.

  2. **Private Key:**

     It’s the key to the public key, it’s forbidden for anyone but you to see.

- **How to create and use a SSH Key?**

  1. **Generate a public key:**

     We can use this command line in the git bash terminal:

     - ssh-keygen -t ed25519 -C "agaballah223@gmail.com"

     ssh-keygen ====> it’s the tool that generates the key.

     -t ed25519 ====> it identifies the type of the encryption algorithm.

```
$ ssh-keygen -t ed25519 -C "agaballah223@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/galil/.ssh/id_ed25519):
Created directory '/c/Users/galil/.ssh'.
Enter passphrase for "/c/Users/galil/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/galil/.ssh/id_ed25519
Your public key has been saved in /c/Users/galil/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:rn2FqiHkjYg/3FpZcWw+WfC0EoociYPWCHNqHoaZb4M agaballah223@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|+.+...  o .      |
|oOoo.o o = .     |
|*+ .o o = +      |
|+o.    = +       |
|E.+ . . S  .     |
| o = = . .. .    |
|....* o .. .     |
| .o... +. .      |
|  oo  o...       |
+----[SHA256]-----+

galil@Galileo MINGW64 /d/ITI/Version Control/Day01/Day1
$ |
```

2. Adding the public key to GitHub account:

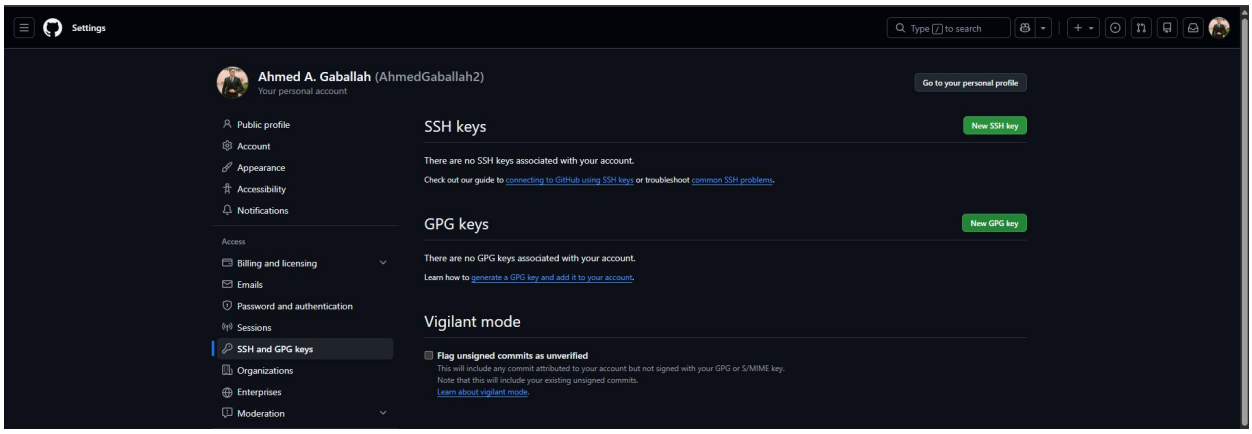
We can use this command line in the git bash terminal:

- cat ~/.ssh/id_ed25519.pub

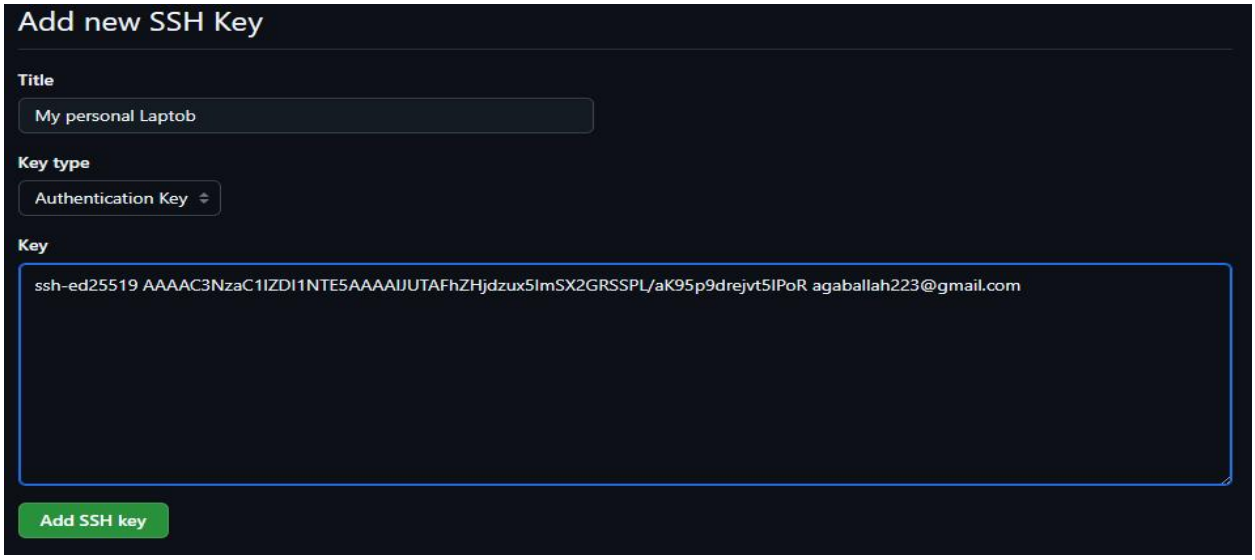
```
galil@Galileo MINGW64 /d/ITI/Version Control/Day01/Day1
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJUTAFhZHjdzux5ImSX2GRSSPL/aK95p9drejvt5IPoR agaballah223@gmail.com

galil@Galileo MINGW64 /d/ITI/Version Control/Day01/Day1
$
```

We can copy the code inside that starts with “ssh-ed25519” and ends with your email address.

Then go to GitHub -> settings -> SSH and GPG keys -> New SSH Key.



Then we paste the code we have copied, and we type a suitable name.



Then we press on Add SSH Key.

3. Try to communicate with GitHub:

We can use this command line in the git bash terminal:

- ssh -T git@github.com

We can check that your account is connected already with GitHub via your public Key.

```
galil@Galileo MINGW64 /d/ITI/Version Control/Day01/Day1
$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enter passphrase for key '/c/Users/galil/.ssh/id_ed25519':
Hi AhmedGaballah2! You've successfully authenticated, but GitHub does not provide shell access.

galil@Galileo MINGW64 /d/ITI/Version Control/Day01/Day1
$ |
```